

MXCuBE status report

Global Phasing

MXCuBE/ISPyB Meeting
Helmholz Zentrum Berlin, 20 June 2026

Brief beamlines summary

- We have no beamlines of our own, however ...
- ... the autoPROC processing software is in use at all MX beamlines, and by numerous end-users, including industrial users
- Global Phasing Data Acquisition Workflow is integrated with MXCuBE.

In various states of readiness at

- EMBL Hamburg P14
 - ESRF MASSIF-1
 - MAX IV BioMAX
 - SOLEIL PX2
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- The Workflow follows the local implementation
 - GPhL code is beamline-agnostic, as it must work at multiple sites
 - The code is kept up to date with the MXCuBE develop branch
 - Site-specific branches are disfavoured
 - We support both Qt and Web interfaces, using pop-up windows specified with a UI-agnostic JSON schema
 - autoPROC runs in stand-alone mode, but results need integration with local LIMS/viewers
 - Multi-sweep experiments and their detailed processing output still go well beyond the capabilities of standard templates for data storage and results presentation
 - Cybersecurity is NMR (Not My Responsibility)
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Workflow installation has been a long-running project.

- EMBL-HH P14: the workflow has been running stably in production mode for five years with over 1000 data sets / year by just one group
- MASSIF-1: the workflow is in production mode for unattended data collection as of *right now*, with the GPhL 'basic' strategy used as a default for standard acquisition, but with more advanced ones available
- SOLEIL PX2: all major obstacles are out of the way, and workflow should reach production-ready for interactive use over the summer
- MAX IV Biomax: Installation is proceeding actively

We plan on moving to a proper versioning/release system for the workflow, to support production use

MXCuBE working groups

- Participate in refactoring, reviewing pull requests etc.
- Started an MXCuBE-Qt Working Group, and working (for now with SOLEIL) to update the Qt version to catch up with the (breaking) Abstract Diffractometer change
- Working with the Automation WG to model the VolumeScan (Mesh scan and X-ray centring), interfacing with MXLIMS model.

- MXLIMS is a data model, file format and interface for sample shipment, experiments and processing results, intended to standardise interactions between users and synchrotrons
 - The basic MXLIMS model is now stable, and work continues in a coalition of the willing
 - Several groups can now produce Sample Shipment documents, and we are collaborating to test ingestion at synchrotrons
 - Collaboration on testing and integration is highly desired
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- “*Automated workflows for strategy computation and data collection at synchrotron beamlines*”, Fogh et al. (2026), Acta Cryst D**82** <https://doi.org/10.1107/S2059798326001920>.
 - Detailed description of workflow program, how strategies are generated and executed, and dose and transmission handling
 - “*Radiation damage in sub-Ångström resolution macromolecular crystallography: a low-dose study*”, Bourenkov et al. (2026)), Acta Cryst D**82**, <https://doi.org/10.1107/S205979832600269X>
 - Radiation damage observed at 0.54Å resolution between two datasets, each collected with a 50kGy dose, separated by a 1MGy irradiation.
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